Track B Project Proposal Template

1 Proposed Project Title and Biological Significance

Give a descriptive title. In 2–4 sentences, explain the biological or biomedical problem and why it matters.

2 Research Question and Prediction Task

State the research question. Identify the target/outcome, its units or classes, and the predictors. Indicate regression, classification, or both. Explain what one sample represents.

3 Dataset and Access

Give the source or citation, estimated number of samples and predictors, variable types, missingness, outcome distribution/class balance, and known grouping or temporal structure. Confirm permission to use the data; do not include restricted data.

4 Proposed Algorithms and Baseline

List at least three course algorithms appropriate for the task, including a simple baseline when applicable. Briefly justify each choice. Any exception requires approval.

5 Validation, Tuning, and Evaluation Plan

Specify how the dataset will be divided into development and independent held-out test sets (normally 80/20). An independent held-out test set is mandatory. Describe the cross-validation strategy (normally 5-fold), any grouping/time-aware adjustments, preprocessing pipelines, primary selection metric, additional metrics, and proposed hyperparameters. Explain how data leakage will be avoided.

6 Complexity Experiments and Biological Interpretation

Describe two planned experiments about overfitting, complexity, or regularization. State how important predictors will be interpreted biologically.

7 Feasibility and Anticipated Challenges

Identify risks such as limited samples, high dimensionality, missingness, class imbalance, compute limits, or data restrictions. Explain why the work is feasible in three weeks and what outputs you expect.

8 Proposal Grading (10 points; completed by instructor)

Criterion	Points
Biological background and research question	2
Dataset description and suitability	2
Selection and justification of algorithms	2
Preprocessing, cross-validation, and evaluation plan	3
Feasibility within three weeks	1
Total	**10**

9 Review

Approval checklist:

☐ Biological relevance
☐ Suitable data with a valid independent held-out test set
☐ Appropriate algorithms
☐ Valid evaluation design
☐ Feasible scope and timeline

Conclusion:

☐ Approved
☐ Approved with revisions
☐ Revise and resubmit

Comments / required changes: